

My FlyRank Project — One-Page Revision Sheet

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IF I REMEMBER ONE SENTENCE

I built a model that finds pages people already see in Google but don't click, and puts them at the top of the editor's to-do list.

THE STORY IN SIX STEPS — SAY THEM IN THIS ORDER

- 1 **The problem**
FlyRank publishes content into client websites. Someone has to decide which pages are worth an editor's time. Right now that comes from hand-written rules that work, but run out when the signals get tangled. **My question: which visible pages under-capture clicks, and who should the editor open first?**
- 2 **The simple rule (the baseline) — built FIRST, then frozen**
If a page is showing in search but its click rate is low → flag it, give it a score, attach the reason code **low_ctr_visible**, and tell the editor to review its title and meta description. Queue of 30,616 pages. **It had to be frozen before the model, so the model couldn't be judged against a target chosen afterwards.**
- 3 **The model**
Inputs from Jan + Feb, answer from March, so it never sees the future. Three models compared on identical data: accuracy goes down as human readability goes up. It outputs a **probability**, and I sort by it — I don't rank by CTR.
- 4 **Grouped by client — why the honest score is lower**
Pages are grouped so the model is only ever tested on clients it has **never seen**. If you shuffle instead, it studies and is tested on the same client, memorises the customer, and the score jumps to 0.836. **That higher score is a lie** — a brand-new client would never be familiar.
- 5 **The audit — the part that makes this project stand out**
I had been saying the honest split "cost me 0.052". Wrong. The base rate also fell (0.246 → 0.216), so **0.030 of that gap — 57% — is the exam getting harder, not my model getting worse**. As a multiple of the base rate the honest split is actually ahead: **3.63x vs 3.40x**. I corrected it in the notebook, on my live site and in the paper.
- 6 **The playbook, then the paper**
Turned the list into a Monday-morning instruction sheet: seven page types, each with an action, an owner and an effort. Best part is what it sets **aside** — 20,061 pages (half) marked watch-only, so nobody wastes time on them. Then wrote it up as a research paper, published on my portfolio site.

THE NUMBERS

OUT OF 50 PAGES GIVEN TO AN EDITOR	RIGHT	SCORE
Random guessing (base rate)	11	0.216
The frozen rule	25	0.492
My model (HistGradientBoosting)	39	0.784

- Beat the rule in **all five folds**, not just on average.
- Random Forest **0.760** — lowest wobble, and readable. Depth-3 tree **0.692**, prints on one page.
- Remove click history and it falls **0.784 → 0.604**. Half the win is borrowed.
- Data: **9.84M** page-day rows · **40,152** pages · **24** clients.

THE TWO SCORES — DON'T MIX THEM UP

R² = regression score

"How close is my predicted number to the real number?" Used **once**, in Week 3, where the real number was **the page's actual CTR**. It was only a leakage test: R² 0.037 → **0.874** the moment I added the click count (the top half of the CTR formula). I removed it and never used regression again.

precision@50 = my real score

"Of the 50 pages I picked, how many were actually right?" My model is yes/no, so there is **no actual number** to be close to. **39 out of 50 = 0.784**.

TRAPS

- Don't say** "it shows how to improve CTR." **Say** it ranks which page to open first — **nothing here is causal**.
- Don't say** "pages with high CTR." **Say** pages earning **far less than pages in that same position normally get**.
- Don't say** "second-page results come first." **Say** my top picks were mostly **page-1 and top-3** pages — a buried page isn't wasting anything.
- Don't say** "grouping saves time." **Say** grouping stops the score being **fake**.

HOW TO CLOSE

An editor handed 50 pages now finds 39 real problems instead of 25 — and the number I'm most confident about is the one I had to correct downwards.

The Results Table — how to read it

Page 2 of 2 · every number below is on the honest, grouped-by-client split

Method	precision@20	precision@50
Base rate (random ordering)	0.216	0.216
The frozen rule (baseline)	0.560	0.492
Logistic regression	—	0.604
Decision tree, depth 3	—	0.692
Random forest	—	0.760
Gradient boosting	0.850	0.784

WHAT THE TWO COLUMNS MEAN

precision@20 — a short queue

Hand the editor only the top 20 pages. Gradient boosting gets 0.850 — that is 17 right out of 20, 3 wasted.

precision@50 — a longer queue

Hand over the top 50. Gradient boosting gets 0.784 — 39 right out of 50, 11 wasted.

THE RULE OF THUMB

- A shorter queue is a more accurate queue. The model is most confident about its very top picks, so the further down the list you go, the more mistakes creep in.
- So the choice of K is a business decision, not a technical one: how many hours does the editor have, and how many wasted trips are acceptable?
- The full curve runs 0.900 / 0.950 / 0.967 / 0.940 / 0.830 / 0.820 at K = 10 / 20 / 30 / 50 / 100 / 200 in the Week 7 playbook — measured on the pooled queue, which is a different (easier) count than this table. Never compare the two directly.

THREE THINGS AN EXAMINER MAY ASK

- Q Why is the base rate 0.216 in both columns?
Because the base rate is the share of under-capturers in the whole test set. It does not depend on how many pages you hand over, so queue length cannot change it.
- Q Why are there dashes in the precision@20 column?
Only the winner and the two reference points (base rate, frozen rule) were run at K = 20. The three middle models were compared at K = 50 only — the headline metric agreed in advance.
- Q The rule scores 0.560 at K = 20 but 0.492 at K = 50 — why does it drop too?
Same reason as the model: its best guesses sit at the top. Both degrade with queue length, and the model stays ahead at both.
- SAY THIS ABOUT THE TABLE
- Every row sat the same exam — same rows, same folds, same metric, same tie policy — and the ordering is the story: simple models score lower, and gradient boosting wins at both queue lengths. I reported all five candidates, not only the winner.